

Lorenzo Di Napoli

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Personal Introduction

Student of Computer Science and Engineering, pursuing a double master's degree in **28Digital Master School | Intelligent Chips and Systems**, studing at the Universities of *Politecnico di Milano* and *Tampere University*.

I am interested in embedded systems and robotics, and I developed various skills in low-level programming; my focus is on understanding systems deeply and the influence of hardware architecture on software applications.

Educational Path

2026 - Present: 28Digital Master School | Intelligent Chips and Systems (Computer Science and Engineering), Politecnico di Milano (IT) and Tampere University (FIN)

2023 - 2026: Bachelor's Degree in Engineering of Computing Systems, Politecnico di Milano (MI) — 100/110L

2018 - 2023: High School Diploma, in Liceo Scientifico opzione Scienze Applicate, Istituto Salesiano Sant'Ambrogio, Milano (MI).

Language Skills

Italian: Native speaker.

English: Level C1, Grade C (Cambridge CAE Certificate).

Soft Skills

- Effective Communication and Teamwork
- Management of Emergencies and Decision-Making under Pressure
- Accountability for own and others' safety and legal responsibilities
- Supervision and Management of Customer Safety
- First Aid and Defibrillator (BLSD) Certification

Technical Skills

Programming Languages

C/C++	● ● ● ● ○	**Java**	● ● ● ○ ○	**Python**	● ● ○ ○ ○
Cuda	● ● ○ ○ ○	**Assembly**	● ● ● ○ ○	**VHDL**	● ● ● ○ ○
HTML & CSS	● ● ○ ○ ○	**JavaScript**	● ● ○ ○ ○		

Tools and Support Technologies

- Linux (Ubuntu, Fedora)
- Makefile · CMake
- Docker/Podman
- ROS2 (Humble)
- Git · GitHub
- Valgrind · Cachegrind · Massif · GDB
- Tomcat · JavaEE Servlets · Thymeleaf · NodeJS
- Vivado · Vitis

Technical Knowledge from University Courses

Computer Architectures (RISC-V) · Digital Logic Design · Operating Systems (Linux)
Algorithms and Principles of Computer Science · Software Engineering
Database and Information Systems · Web Technologies (Tomcat, Thymeleaf, NodeJS)
Basic Circuit Theory · Introduction to Electronics
Fundamentals of Automatic Control · Industrial Automation · Robotics (ROS2 - Humble)

Economics & Business Administration

GPU Architecture (CUDA) · FPGA Architecture (VHDL & HLS)

Projects

University Projects

Movhex: development in **C** of a program that allows creating and editing a 2D hexagonal map, with the objective of finding the lowest-cost path between two points.
The final result reached a temporal complexity of $O(n \log(n))$ and spatial complexity of $O(n)$; where n is the number of hexagons present in the map.

Priority-List: design and implementation in **VHDL** of a HW module, which performs the function of a controller for a Priority-List of Tasks.
The final result was implemented via a fully synthesizable FSM on an FPGA, without Latches of any kind.

Mesos: design, implementation, testing and deployment of an Online Game-App based on the board game "Mesos", fully realized in **JavaEE** and distributed via **jar** packages.
The full Project was realized by a team of 4 people, and its implementation supports *TUI/GUI* layouts, and *Socket/RMI* communication protocols.

Personal Projects

Home Server: Self-hosted personal server running various services (including *Nextcloud* and *Immich*) in rootless containers, each one isolated with its own dedicated UID/GID range.

Services are reachable only over a *WireGuard* VPN, with *Caddy* handling HTTPS reverse-proxying for distributed architectures and *dnsmasq* providing local DNS.

Currently building a dedicated backup NAS: a 5-drive RAID 6 array managed via *mdadm* software RAID, behind an HBA configured in IT mode.

LDN Library: personal general-purpose library implemented in **C++20** using the advanced features of the language.

Neural Networks: from-scratch implementation of different neural networks in **Python** and **Jupyter Notebook**, focused on clustering various geometric shapes (circles, spirals, etc.).

Work Experiences

Lifeguard

First aid and defibrillator (BLSD) certification, civil and criminal liability for one's actions and those of swimmers.

- **June 2025 - December 2025 [116 hours]:** Summer and winter work experience as Lifeguard Assistant P at the sports center *"M.G.M.Sport Srl"* in Paderno Dugnano (MI).
- **June 2024 - July 2024 [171 hours]:** Summer work experience as Lifeguard Assistant P at the sports center *"M.G.M.Sport Srl"* in Paderno Dugnano (MI).
- **July 2022 [36.5 hours]:** Summer work experience as Lifeguard Assistant P at the sports center *"M.G.M. Sport Srl"* in Desio (MB).

Certifications and Documents —> DOWNLOAD

Bachelor's Degree: awarded in Computer Science and Engineering, final grade 100/110L — [29/09/2026]

Cambridge CAE Certificate: English language certification, level C1 Grade C — [09/01/2026]

Driver's License: Category B driver's license — [18/11/2023]

Lifeguard Certification: award of the Lifeguard Assistant certificate (levels P, IP, MIP) at the *"Federazione Italiana Nuoto" (FIN)* — [09/11/2022]

SFA Certificate: certificate of participation and completion of the *"Scuola Formazione Animatori MGS" (SFA)* at the association *"Movimento Giovanile Salesiano" (MGS)* — [17/04/2023]